Reliability of AI-Based Visual Inspection for Energy and Industrial Assets: A Systematic Scoping Review and Evidence Map

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***Abstract*—AI-based visual inspection is increasingly proposed for energy and industrial assets, but inspection decisions depend on more than benchmark accuracy: models face distribution shift, unreliable labels, failing sensors, and operators who must act on their outputs. This provisional systematic scoping review is structured using PRISMA-ScR and examines empirical studies that propose or evaluate a reliability property of learned image-based inspection models in industrial, energy or infrastructure settings. OpenAlex searches, a Crossref pilot and seed records, supplemented by citation tracing executed on 26 October 2026 with a publication cutoff of 22 October, yielded 6,970 unique reports; 6,269 have title/abstract dispositions and 701 remain pending. The assessed full-text subset comprises 157 provisionally included studies charted from 157 primary reports, with 13 associated secondary reports. The remaining search additions have not yet completed selection and retrieval. Evidence has grown quickly (67% of studies from 2025–2026) but is concentrated: 99 studies address distribution shift or robustness, whereas 16 address selective or risk-controlled decisions, 12 multimodal or sensor failure, and only 4 provide finite-sample decision guarantees. Most studies (62%) rely on public benchmarks, 4 report deployed or operational use, and 7 evaluate decisions with human inspectors. Energy-asset contexts account for 21 studies (13%). Studies that test realistic conditions repeatedly report large performance losses, benchmark saturation, leakage and threshold sensitivity that standard metrics hide. Within the assessed subset, the findings motivate moving reliability evaluation from model scores to inspection decisions: natural-shift test sets, operating-point and coverage metrics, explicit referral policies with measured workload, and field validation in energy settings. The review is limited by a single bibliographic index, open-access-only retrieval (568 reports not retrieved) and AI-assisted single-reviewer screening with unresolved audit documentation and incomplete screening of citation-tracing additions.**

***Index Terms*—Industrial inspection, energy assets, trustworthy artificial intelligence, visual anomaly detection, distribution shift, uncertainty, selective prediction, human-in-the-loop, scoping review, evidence map.**

I. Introduction

A. Industrial Context and Motivation

Visual inspection supports decisions on energy and industrial assets: whether a photovoltaic (PV) module is defective, whether a transmission-line insulator needs replacement, whether a weld, casting or printed circuit board can be shipped, and whether corrosion on a pipe or bridge needs intervention. Learned image models now detect, segment and classify such defects from RGB, thermal, electroluminescence (EL), X-ray and 3D data. An image-based finding is, however, an observation of the surface or of an imaging response, not a direct measurement of physical condition, remaining life or fitness for service. An inspection decision built on it inherits errors from acquisition, from the model and from the threshold that converts a score into an action.

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B. Reliability Challenges in Visual Inspection

Several properties beyond average accuracy determine whether such a decision can be trusted: robustness when lighting, viewpoint, product or site change; uncertainty estimates and calibrated confidence; principled thresholds, abstention and referral; tolerance of noisy labels and contaminated training data; behaviour when a sensor or modality fails; grounding of generated explanations and reports in image evidence; and evaluation designs that reflect deployment rather than curated benchmarks. We treat these as dimensions to be mapped, not as established field-wide gaps.

C. Existing Reviews and the Need for This Evidence Map

Recent surveys organise industrial anomaly and defect detection by learning paradigm, architecture, modality or dataset [1]–[6]. A foundation-model survey frames cold-start inspection around target-data availability and cross-dataset protocols [7], and a PRISMA-informed review addresses the simulation-to-field gap for underground inspection [8]. Table I compares them, based on reading their full texts. Shift and evaluation realism recur as challenges, and the comparison motivated mapping reliability dimensions across industrial and energy visual inspection together with report-level selection records and decision-level variables (guarantees, referral, human evaluation and validation maturity). This review addresses that combination; it does not claim to be the first review of industrial visual anomaly detection.

D. Review Objectives, Questions and Contributions

The objective is to map the empirical evidence on reliability of learned visual inspection models for energy and industrial assets. Four questions guided the review:

RQ1 In which asset contexts, tasks and imaging modalities has reliability been studied?

TABLE I
COMPARISON WITH RETRIEVED PRIOR REVIEWS (CHARACTERISED FROM THEIR FULL TEXTS).

Review	Scope	Search and selection	Reliability coverage	Distinction of this review
[1]	Deep industrial image AD: architectures, supervision, losses, metrics, datasets	Narrative; no reproducible search reported	Industrial settings (few-shot, noisy, continual) discussed as open settings	Maps evidence on decision reliability, not methods
[2]	Industrial visual AD across five learning strategies, 196 papers, 2D/3D datasets	"Systematically review" without search strings or flow	Generalisation and benchmark results; little on calibration, referral or humans	Report-level selection; R1–R8 charting
[3]	Visual AD across domains by sample number, modality, hierarchy	Narrative survey	Few-shot and modality challenges; reliability not a focus	Industrial/energy focus; deployment variables
[4]	Vision-based industrial AD pipeline, 2019–2023	Searches of IEEE Xplore, Scopus, Springer, Taylor & Francis	Acquisition, evaluation and operator issues; shift briefly	Extends to 2026; decision guarantees, human evaluation
[5]	RGB, 3D and multimodal unsupervised industrial AD	Narrative classification	Practical challenges and multimodal fusion; sensor failure rarely	Sensor failure and missing-modality evidence mapped
[6]	Image-based quality control, >40 methods with a benchmark	Narrative plus own benchmark	Datasets and metrics as limiting factors	Broader reliability scope and field evidence
[7]	Foundation models for cold-start industrial AD	Narrative; deployment-oriented formulation	Data assumptions, shift, contamination of pretraining	Covers non-foundation methods and decision level
[8]	Simulation-to-field gap in underground mine inspection	PRISMA-informed, six databases, 2015–2026	Sim-to-real, domain adaptation, field validation	Industrial/energy assets; all eight dimensions

RQ2 Which reliability dimensions are proposed or evaluated, and with which data and metrics?
RQ3 How are inspection decisions formed: thresholds, abstention, referral, guarantees and human review?
RQ4 How mature is the validation (benchmark, industrial data, field, deployment), and which evidence is missing?

The contributions are (i) a reproducible, report-level evidence base of 157 studies with open selection and charting records; (ii) an evidence map across eight reliability dimensions, contexts and validation maturity; (iii) a synthesis of decision-level evidence (guarantees, referral, workload, human evaluation); and (iv) research opportunities qualified by their evidential basis.

E. Organization of the Paper

Section II defines the concepts; Section III reports the methods; Sections IV and V report results; Section VI discusses them and Section VII concludes.

II. Background and Conceptual Scope

A. Visual Inspection Tasks and Modalities

Eligible tasks are classification, detection, segmentation, anomaly detection (image or pixel level), measurement from images, and description or question answering about inspection images. Eligible modalities are image-based sensing: RGB, thermal infrared, EL, multispectral or hyperspectral, X-ray radiography and computed tomography (CT), depth and point clouds, video, and multimodal combinations that include imagery. Core contexts are energy assets (generation, transmission, PV, wind, nuclear, oil and gas, process plants) and manufactured products; transferable infrastructure (bridges, roads, rail, tunnels, buildings) is labelled adjacent. Imaging non-destructive testing outside these modalities, generic methods without industrial evaluation, and non-visual condition monitoring were treated as background unless they contributed a transferable reliability method.

B. Reliability Dimensions and Decision Units

We chart eight dimensions. R1 covers uncertainty quantification, confidence and probability calibration; calibration is coded only when calibration of predicted probabilities is measured or targeted. R2 covers selective prediction, abstention, rejection and risk-controlled or conformal decisions. R3 covers distribution shift, domain adaptation and generalisation, out-of-distribution (OOD) and open-set detection, and robustness to acquisition, corruption or adversarial perturbation. Zero- and few-shot transfer counts as R3 only when evaluated as a shift. R4 covers label noise, annotation reliability and training-set contamination. R5 covers human-in-the-loop designs, referral and human–AI decision processes. R6 covers multimodal reliability, missing modalities and sensor failure or degradation. R7 covers grounded or evidence-linked outputs, including hallucination of vision-language models (VLMs). R8 covers evaluation validity: dataset realism, saturation, leakage, shortcut learning and benchmark design. Decisions are made at different units (pixel, image, object, part, asset, lot) and at different operating points, so we chart decision policies and guarantees separately from detection scores.

C. Human Review, Multimodal Evidence and Grounding

Human review appears as a recommendation, as a designed referral path, or as an evaluated human–AI process; we distinguish the three. Multimodal fusion that improves accuracy is distinguished from evidence of reliability when a modality is missing or degraded. Explanations and generated reports

TABLE II
ELIGIBILITY RULES (PROTOCOL V0.4–V0.5).

Dimension	Inclusion rule	Exclusion code / handling
Context	Energy, industrial or infrastructure assets; manufactured products	TA01/FT01 (surveillance, driving, medical-only, generic)
Modality	Image-based sensing incl. thermal, EL, X-ray/CT, 3D, video, multimodal with images	TA02/FT02; imaging NDT outside list as background
AI task	Learned model performing an inspection task	TA03/FT02
Reliability	Proposes, evaluates or benchmarks R1–R8	TA05/FT06 (accuracy-only or asserted robustness)
Report type	Empirical studies, benchmarks, theses, preprints	TA04/FT03 (non-empirical, talks, front matter)
Versions	Verified versions linked to one study	FT04 for secondary versions
Language	English full text	FT05 (reported)
Dates	No lower limit; searches to 16–22 Oct. 2026	–

are distinguished from validated grounding, faithfulness or usefulness.

III. REVIEW METHODOLOGY

A. Review Design and Reporting Framework

A systematic scoping review with an evidence map suits questions about the extent, nature and gaps of a heterogeneous, fast-moving literature, where pooled effect estimates are not meaningful. Conduct followed JBI guidance for scoping reviews [9]; reporting follows PRISMA-ScR [10], and search reporting follows PRISMA-S [11].

B. Protocol, Registration and Amendments

The protocol was written before screening (version 0.3, 13–14 October 2026) and amended with dated entries: version 0.4 (16 October 2026) refined eligibility into the R1–R8 core criterion, added the background/supporting category and the OpenAlex search, and recorded the delegated screening arrangement; version 0.5 (22 October 2026) recorded the screening outcome, version linkage and the full-text procedure. The protocol was not registered. All versions, amendments and decision records are in the project repository.

C. Eligibility Criteria

Table II summarises the criteria. All four core conditions were required: context, image-based modality, a learned model performing an inspection task, and at least one reliability dimension that the study proposes, evaluates or benchmarks. A generic robustness claim without a corresponding design or evaluation did not qualify (code FT06). Reviews, methodology papers, foundational benchmarks without a reliability dimension, and adjacent non-visual studies contributing transferable reliability methods were charted as background, not as core evidence.

D. Information Sources and Search Strategy

Scopus, Web of Science and IEEE Xplore exports were not available to the authors. OpenAlex, which indexes Crossref, arXiv, institutional repositories and publisher metadata, served as the multidisciplinary index (Table III). Three Boolean title/abstract families crossed context, imaging, AI and reliability vocabulary (R1/R2; R3; R5–R7). A review-overlap family targeted surveys. Because OpenAlex lacks abstracts for some publisher records, a title-only family searched records without abstracts. A title-restricted generic reliability family and a citation family (works citing MVTec AD [12] with reliability vocabulary) complemented them. A capped Crossref pilot (16 query families, 13 October 2026) and 33 seed records were retained as identification sources. Search development is reported: one run failed on pagination, one was superseded after a known-item test showed missed core records (blocks were revised), and one failed on a rate limit; the definitive runs completed on 22 October 2026. Exact strings, fields, filters, raw responses and checksums are in the repository (Appendix A). Of the ten primary seed studies, seven were retrieved by the definitive OpenAlex runs. The three missed preprints were recovered from their arXiv identifiers. This partial known-item recall is reported as a limitation. An initial citation-tracing attempt on 23 October failed; the subsequent completed run OPENALEX-20261026-CT02 was integrated. It added 1,943 retrieval occurrences and 1,385 previously unregistered reports. Of 850 previously saved AI decision proposals for these additions, 664 were reconciled on 2 November with their original reviewer/date; 186 title-only exclusion proposals were held pending for source assessment. Screening and retrieval of the expanded corpus remain incomplete (Section VI-F).

E. Record Management, Deduplication and Study Linkage

Identification events (one per returned record), deduplicated report records and studies are kept as separate units. Records were deduplicated by DOI, with arXiv DOIs normalised to arXiv identifiers; records with different identifiers were never merged on title. Same-title pairs (209) were verified on order-insensitive author lists, year, record type, venue and identifiers: 144 links were verified and grouped into study identifiers, 25 pairs were rejected as distinct works or spurious non-Latin-title matches, and 40 remained unverified for lack of metadata. Every count in this paper is regenerated from these records by scripts.

F. Selection and Reviewer Arrangement

The first author screened a 20-record calibration set and then delegated title/abstract and full-text eligibility judgement to an AI assistant (Claude, Anthropic) operating under the protocol, with a written rationale and an evidence locator for every decision. Fourteen dated screening notes fixed consistency rules (for example, the zero-shot, calibration and imaging-NDT rules above). Unclear records proceeded to full text. Full texts were retrieved only from open-access locations (OpenAlex locations, a DOI lookup and exact-title arXiv

TABLE III
EXECUTED SOURCES AND SEARCH PROVENANCE.

Source/platform	Date and fields	Families and returned records	Restrictions and coverage	Run identifier
OpenAlex REST API	22 Oct. 2026; title and abstract	R1/R2 1,459; R3 932; R5–R7 842; review overlap 640	Published to 16 Oct. 2026; all pages retrieved	OPENALEX-20261022-01
OpenAlex REST API	22 Oct. 2026; title only	No-abstract records 339	has_abstract:false	OPENALEX-20261022-01
OpenAlex REST API	22 Oct. 2026; title / citation	Generic reliability in titles 976; citing MVTec AD 1,062	cites:W2948982773	OPENALEX-20261022-02
OpenAlex citation tracing	26 Oct. 2026; eight forward/backward families	1,943 additional retrieval occurrences	Publication cutoff 22 Oct. 2026; selection unfinished	OPENALEX-20261026-CT02
Crossref REST API	13 Oct. 2026; bibliographic query	16 capped pilot families, 400 occurrences	Ranked, capped; supplementary	PILOT-20261013-01/02
Seed records	13 Oct. 2026	33 known studies, benchmarks, reviews, guidance	Metadata verified against DOI/arXiv	SEED-20261013

search), with every attempt logged. Reports not retrieved were not excluded. Full-text decisions cite page locators and were made per study on the most complete retrieved report. This is AI-assisted single-reviewer selection, not independent dual review. A human verification audit of all inclusions and a random 10% of title/abstract exclusions is planned before submission.
[Author input required: report the completed audit: auditor, sample, agreement and resolution of disagreements.]

G. Data Charting

A charting form operationalised the R1–R8 dimensions and decision variables. It was piloted on ten studies chosen to span dimensions, contexts and maturity, then revised (version 0.2) to add shift origin (natural, synthetic, both), guarantee type (finite-sample, empirical operating point, none), human role (none, recommended, designed into workflow, evaluated with humans) and controlled vocabularies for context and validation maturity. Every charted field carries a page locator. "Not reported" means the full text was read and the item was absent; "Not extracted" means it was not charted. Charting was performed by the same AI assistant and is subject to the same audit. Reports that could not be retrieved were coded at abstract level only (reliability dimensions from the screening rationale) in a separate tier (`charting/abstract-tier.csv`), retained as an exploratory screening-rationale summary, not an independently charted sensitivity analysis. An abstract-capture flag does not verify full-text eligibility.

H. Assessment of Evaluation and Evidence Limitations

Consistent with scoping-review practice, no risk-of-bias score was assigned. Instead, evaluation features that bear on reliability were charted: shift origin, data access, guarantees, decision policy, human involvement and validation maturity. Leakage, contamination and saturation were charted where studies reported them (R8).

I. Evidence Mapping and Synthesis Methods

Counts are of studies; dimensions are multi-label, with the primary dimension listed first. Denominators are stated with each count. Synthesis is descriptive and narrative: we group findings by dimension and report the authors' own results with attribution, without pooling incomparable scores.

J. Reproducibility and AI Assistance

The protocol, queries, raw search responses with checksums, identification events, decision packets, linkage, retrieval logs, charting records and the scripts that produce every figure and count are released in the project repository; copyrighted full texts are not redistributed. AI assistance was used for screening, charting and drafting under the first author's delegation; software validation of these records is not a substitute for human review.

IV. Results: Study Selection and Evidence Characteristics

A. Search and Selection Outcomes

Searches produced 8,626 identification events (OpenAlex 8,193; Crossref pilot 400; seeds 33). Removing 1,656 duplicate occurrences leaves 6,970 report records. At this revision, 6,269 have recorded title/abstract dispositions and 701 remain pending (Fig. 1). Screening has excluded 4,953 and assigned 463 to background. Of 853 reports routed to full text or retained as unclear, retrieval was initiated for 787; 66 additional reports await retrieval initiation. Among the attempted reports, 568 have not been retrieved and 219 were assessed from full text. These yielded 157 primary included reports representing 157 provisionally included studies; 13 linked secondary reports are associated with those studies. The full-text exclusions comprise 36 substantive/language exclusions and 15 secondary-version exclusions; 11 reports were background. The evidence map below describes only the assessed subset and is not the final selection outcome of the expanded search.

B. Publication and Study Characteristics

The evidence is recent: 105 of 157 studies (67%) appeared in 2025–2026 (Fig. 2). The 2026 count covers January to October only. Preprints, conference papers and theses are common, reflecting the pace of the field; versions were linked to avoid double counting.

C. Assets, Contexts, Tasks and Modalities

Manufactured products dominate (105 studies), followed by adjacent infrastructure (27) and energy assets (21; Table IV), with 4 in other adjacent settings. Energy studies

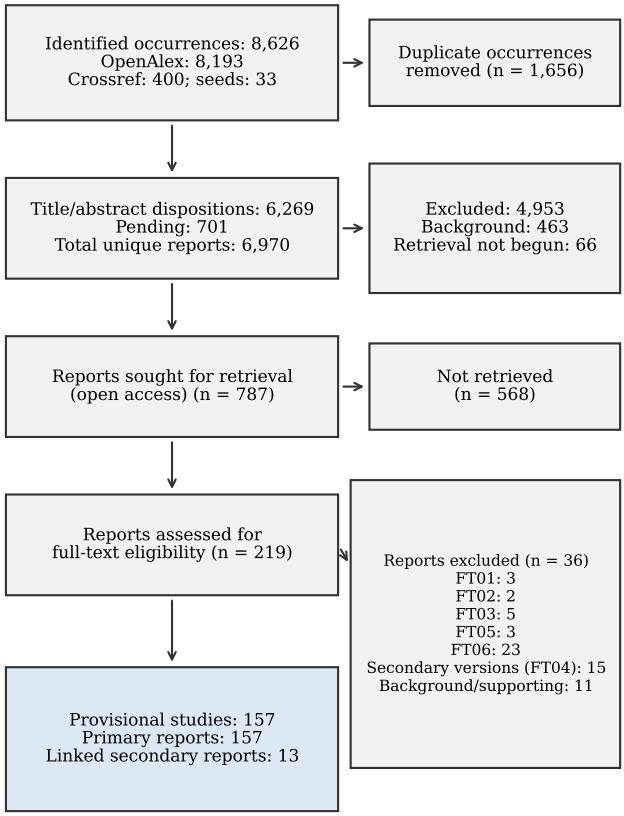


Fig. 1. Selection flow adapted from PRISMA-ScR. FT01–FT06: full-text exclusion codes (Table II). Records not retrieved were not excluded; they await retrieval or abstract-level charting.

cover PV modules and PV glass, concentrated solar power receivers, transmission-line insulators, substation infrared sensors, railway catenary components, nuclear plant components, chemical-plant piping, analog gauges and drone inspection of wind and solar assets. RGB imaging dominates. Thermal, EL, X-ray/CT, 3D and multimodal inputs are a minority, and anomaly detection on normal-only training data is the most common task formulation.

D. Datasets, Partitions and Evaluation Settings

48 studies use the MVTec AD family and 19 use VisA. 24 rely at least partly on proprietary or restricted data. Code or data release was stated in 41 studies. Threshold-free ranking metrics dominate: AUROC-type metrics appear in 53 reliability evaluations, against 12 that measure calibration, 5 that report risk–coverage or coverage, and 7 that report false-alarm rates at an operating point.

V. Results: Reliability Evidence Map

Fig. 3 shows the distribution of dimensions across contexts: R3 99, R8 50, R1 43, R4 29, R5 25, R7 24, R2 16 and R6 12 studies.

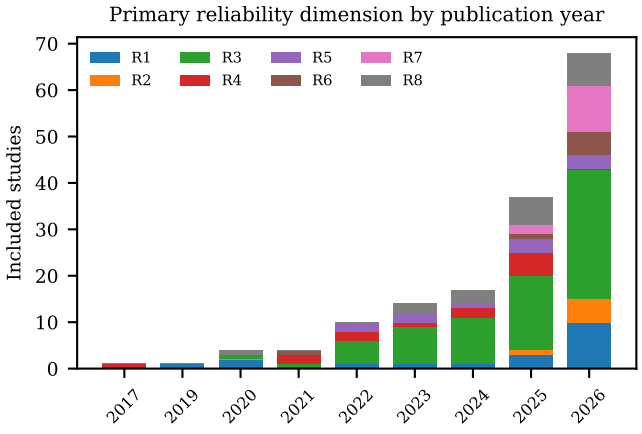


Fig. 2. Included studies by publication year and primary reliability dimension (n = 157). The 2026 bar covers the search period to October.

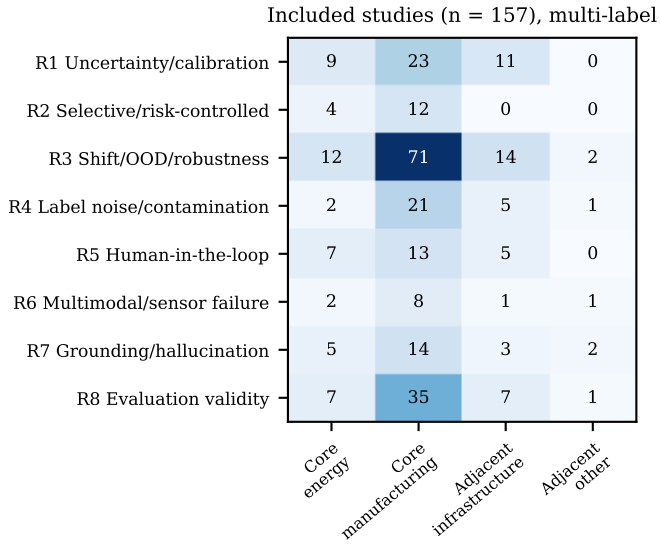


Fig. 3. Evidence map: studies addressing each reliability dimension by context (multi-label; n = 157).

A. Robustness and Distribution Shift (R3)

Shift is the most studied dimension [13], [15], [16], [18]–[25], [30], [34]–[120]. Of the R3 studies, 71 test natural shifts (alone or with synthetic ones) and 26 test synthetic shifts only. Studies with natural shifts consistently report large losses. A crack classifier fell from 94.1% accuracy on its benchmark to 14.3% on field images and recovered after fine-tuning on 380 field images [36]. State-of-the-art detectors that saturate MVTec AD degraded under within-category background, illumination and viewpoint shift [93], [118], under robot-captured viewpoints and reflectance [42], under outdoor maintenance scenes [25] and under railway LiDAR sensor changes [43]. Corruption benchmarks quantify sensitivity to noise, blur and illumination [97], [100]. Adversarial attacks collapse near-perfect clean scores [98], [101], [104], and poisoned training data can make defect inspectors misjudge criticality or miss defects [50]. Mitigations include domain adaptation from lab-

TABLE IV
INCLUDED STUDIES IN ENERGY-ASSET CONTEXTS (N = 21). DIMENSIONS ARE LISTED PRIMARY FIRST.

Study	Asset/context	Modality	Dimensions	Reliability mechanism (as reported)	Maturity
[13]	Steel/alloy corrosion on assets	RGB	R1; R3	Bayesian variants of HRNetV2 giving aleatoric and epistemic pixel uncertainty	Industrial dataset (offline)
[14]	PV modules	RGB PV panel images	R1; R2; R8	MC-dropout uncertainty with learned-threshold abstention routing ambiguous samples to inspecto...	Benchmark (public)
[15]	PV module manufacturing/inspection	EL/visual PV images	R1; R3	MC-dropout predictive variance and ECE-style calibration loss; variance-based trust threshold	Industrial dataset (offline)
[16]	Pipeline corrosion/erosion	RGB	R1; R3; R8; R5	MC-dropout predictive entropy for triage; zero-shot cross-domain tests; leakage-free CV; ultra...	Industrial dataset (offline)
[17]	Railway catenary contact-wire supports	RGB	R1	Pixel prediction variance aggregated into image uncertainty to reject blurred images; geometri...	Industrial dataset (offline)
[18]	Analog gauges in industrial plants	RGB	R1; R2; R3; R7	Calibration of numeric correctness, risk-coverage curves, leave-one-dataset-out transfer, corr...	Industrial dataset (offline)
[19]	Concentrated solar power receivers	Thermal-IR	R2; R3; R8	Risk-control thresholding with finite-sample guarantees on a chosen risk; abstention deferring...	Deployed or operational
[20]	PV modules	EL and other PV imaging m...	R2; R3; R5; R7	Risk-coverage selective prediction; robustness to realistic corruptions; reasoning grounded in...	Benchmark (public)
[21]	Transmission-line insulators	RGB	R3; R8; R4	Cross-dataset generalisation evaluation; split audit revealing scene-level leakage and correct...	Benchmark (public)
[22]	PV modules	EL	R3	Continual self-evolving learning adapting to data shifts and emerging defect categories	Deployed or operational
[23]	Nuclear power plant components	RGB	R3; R1	Augmentation versus no augmentation under synthetic-to-real shift; analysis of saturated confi...	Industrial dataset (offline)
[24]	PV glass manufacturing	RGB line-scan images	R3; R1	Domain-balanced sampling and domain consistency objective; evaluation of accuracy and calibrat...	Industrial dataset (offline)
[25]	Outdoor maintenance inspection	RGB	R3; R8	Photorealistic MIAD dataset for maintenance inspection under uncontrolled outdoor conditions;...	Benchmark (public)
[26]	Power-grid insulator shells	RGB	R5; R7; R8	Deep SAD on GradCAM/CartoonX explanations flags anomalous explanations for operator reclassifi...	Industrial dataset (offline)
[27]	External pipe corrosion in chemical plants	RGB	R5	Expert tacit knowledge converted to annotation criteria; predictions verified by maintenance p...	Deployed or operational
[28]	External pipe corrosion in chemical plants	RGB	R5; R4	Operational evaluation of the HITL system; subjective visual severity assessment addressed by...	Deployed or operational
[29]	Solar panel farms	RGB	R5	Human-in-the-loop mechanism within the decision framework	Benchmark (public)
[30]	PV systems	Multimodal	R6; R1; R3; R5	Uncertainty gating scales physics priors and modalities when sensors are missing or corrupted;...	Benchmark (public)
[31]	Substation infrared monitoring sensors	Thermal-IR sensor images	R6	Classifies infrared-sensor failure modes from sensor outputs	Industrial dataset (offline)
[32]	Drone inspection of wind, solar and property...	RGB; Thermal-IR	R7; R8	Critical evaluation of unified VLM inspection: grounding, prompt-variability brittleness, dege...	Industrial dataset (offline)
[33]	Wind turbine blades	RGB	R7	Reports grounded in indexed maintenance procedures; hallucination rate, BLEU-4, ROUGE-L and LL...	Benchmark (public)

oratory to field [62], [64], [119], domain generalisation [57], [105], test-time adaptation [85], [94], [115], nuisance filtering and few-shot recalibration of frozen detectors [116], [117], synthetic-to-real training [23], [58], [81], [86], and continual learning. Continual-learning benchmarks under realistic drift found that dedicated methods rarely beat replay and that geometric drift reduced every method to chance [99]. Open-set formulations evaluate detection of unseen defect types [52], [55], [63], [95], [96], [113].

B. Uncertainty and Calibration (R1)

Uncertainty is studied in 43 studies [13]–[18], [23], [24], [30], [39], [40], [43], [46], [47], [67], [82], [87], [89], [90], [109], [116], [121]–[142], most often through Monte Carlo dropout, Bayesian layers or ensembles in segmentation and classification [13], [17], [132], [133], [136]. Studies report that epistemic uncertainty rises on OOD images [13] and that uncertainty correlates with misclassification [133]. Measurement of calibration is less common. Exceptions report expected calibration error for PV defect models [15], propose ground-truth-free calibration metrics for manufacturing inspection [127], correct overconfident crack classifiers [90], evaluate calibration on unseen production domains [24], and report reliability diagrams and risk–coverage curves for VLM gauge reading on industrial data [18]. In the last study, high-confidence errors persisted after fine-tuning. Uncertainty of image-derived measurements (corrosion severity, monocular point clouds, super-resolved CT pores) is a distinct strand [122], [128], [130].

C. Decision Policies and Human Inspection Workload (R2, R5)

Table V lists the 4 studies with finite-sample guarantees, the 16 with empirically set operating points, and those evaluated with humans. Finite-sample guarantees take the form of conformal p-values for NDT anomaly detection [121], risk-controlled thresholds with expert abstention deployed for months at two CSP plants [19], distribution-free false-alarm tolerance limits [143], and exact binomial certificates for selective image restoration [142]. Two of these state that coverage does not extend to batch dependence or domain shift [142], [143]. Empirical policies route high-uncertainty samples to inspectors: rejecting the most uncertain 20% raised PV fault-classification accuracy from 92.2% to 97.2% [14]; reviewing the top 25% by entropy captured most residual corrosion errors [16]; and a three-way pass/review/reject policy

traded workload against error targets [54]. Human involvement is designed into the workflow in 23 studies [14]–[16], [19], [26], [29], [30], [52], [54], [68], [122], [125], [127], [129], [131], [132], [137], [138], [142], [144]–[147] and evaluated with humans in 7 [20], [27], [28], [59], [67], [148], [149]. These include a preregistered field experiment with factory workers, in which heatmap explanations improved task performance [148]; concordance with a certified PV inspector [20]; inter-rater reliability of steel micrograph labels [59]; inter-operator threshold variance [67]; plant operation of a corrosion system [27], [28]; and a field trial of drone-AR bridge inspection [149]. Review workload was quantified in a few studies, for example 15% of insulator images flagged for manual reclassification [26].

D. Label Noise and Training Contamination (R4)

R4 evidence [21], [28], [34], [48], [50], [59], [66], [69], [89], [103], [106], [114], [116], [124], [127], [135], [139], [141], [144], [147], [150]–[158] mostly injects anomalies into normal-only training sets at controlled rates [103], [141], [150], [151], [155], [157], [158]. Fewer studies examine naturally occurring label problems: inconsistent labels of suspicious regions [139], noisy multi-label reports from historical sewer CCTV databases [153], low inter-rater agreement in metallography [59], and a consensus contamination audit that changed model rankings on a manufacturing dataset [144].

E. Multimodal Reliability and Sensor Failures (R6)

Sensor-failure evidence is sparse [30], [31], [35], [41], [60], [83], [88], [99], [140], [159]–[161]. Missing-modality tests show that a perturbation curriculum raised welding-defect macro-F1 by 21.5 points when modalities were dropped [35], that removing 3D input cost 4.6 points of image AUROC in RGB–3D detection [88], and that uncertainty gating degrades gracefully when PV sensor modalities are missing [30]. One energy study diagnoses failures of substation infrared sensors themselves [31]. Others compensate failing vision with tactile or event sensing [41], [160] or filter weld-pool disturbances [140].

F. Evidence Grounding and Traceability (R7)

Grounding evidence [18], [20], [26], [32], [33], [38], [77], [82], [84], [87], [110], [120], [125], [130], [138], [142], [145], [147], [157], [159]–[163] is largely recent and VLM-based (25 studies use foundation or vision-language models). Evaluations of general VLMs report weak grounding. A unified VLM for drone inspection of wind and solar assets underperformed task-specific models and defaulted to memorised answers [32]. Zero-shot multimodal models reached F1 of about 0.42 on building-defect perception [147], and VLMs grounded aerial road damage poorly [162]. Retrieval-augmented and tool-grounded designs reduce hallucination in maintenance reports for wind-turbine blades [33], additive-manufacturing anomaly classification [163], X-ray NDT [138] and cross-domain anomaly detection [87], [110], [159]. Hallucination also arises in reconstruction: super-resolved CT [130] and diffusion-based sparse-view CT [77].

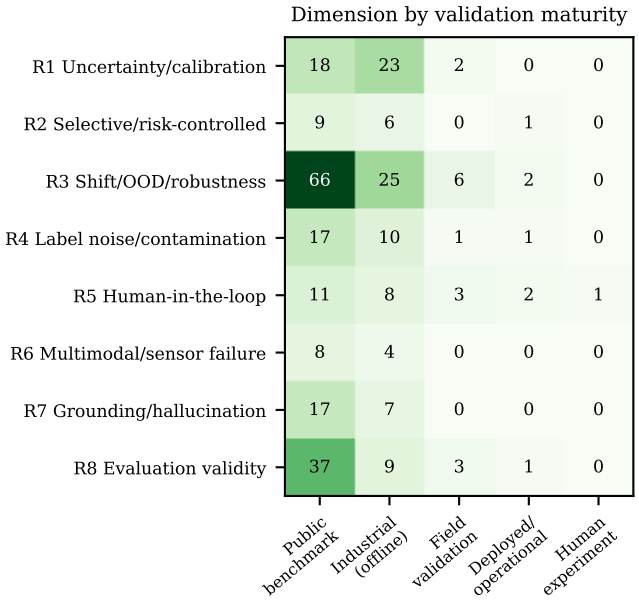


Fig. 4. Coverage matrix: studies by reliability dimension (multi-label) and validation maturity (n = 157). Zero cells mean no included study; they do not establish absence of such evidence in unretrieved reports.

G. Evaluation Validity and Validation Maturity (R8)

R8 studies [14], [16], [19], [21], [25], [26], [32], [34]–[36], [42], [44], [45], [47], [48], [52], [55], [60]–[62], [66], [67], [69], [74], [87], [91], [97], [99], [100], [106], [107], [109], [114], [117], [118], [126], [134], [143], [144], [147], [154], [156], [162]–[169] document the gap between benchmarks and inspection. Saturation motivates new benchmarks: state-of-the-art methods score below 60% AU-PRO on MVTec AD 2 [34], the best challenge entry reached about 57% segmentation F1 on held-out data [61], and rankings shift on real-world and same-sort datasets [48], [69], [114], [154], [156]. Leakage inflates results: a split audit found 41.7% of test images sharing source scenes with training [21], and documentation-derived datasets contain repeated views and annotation errors [66]. Low damage prevalence makes most positive alerts false even for accurate classifiers [134]. Probability-of-detection curves link AI crack detection to engineering inspection reliability [126]. Clever Hans strategies go undetected by standard validation [164], [165], [169], and model selection without labelled anomalies is itself unreliable [166], [167]. Validation maturity is low (Fig. 4): 98 studies rely on public benchmarks, 46 on offline industrial data, 8 on field validation [36], [43], [62], [66], [68], [119], [122], [149] and 4 on deployed or operational use [19], [22], [27], [28]; 1 is a human-subject experiment.

VI. Discussion

A. Answers to the Review Questions

RQ1: reliability evidence is concentrated on RGB anomaly detection for manufactured products. Energy-asset studies (21) are few and mostly offline, the main exceptions being operational studies in CSP plants, PV EL inspection and

TABLE V
STUDIES REPORTING A DECISION GUARANTEE OR AN EVALUATION WITH HUMANS (N = 25).

Study	Context	Decision or referral policy (as reported)	Guarantee	Human role	Maturity
[14]	PV modules	Reject if MC-dropout STD above learned threshold; route to human inspectors	Empirical operating point	Designed into workflow	Benchmark (public)
[15]	PV module manufacturing/inspection	Conservative policy flags high-variance detections for review	Empirical operating point	Designed into workflow	Industrial dataset (offline)
[16]	Pipeline corrosion/erosion	Human review of top-25% entropy	Empirical operating point	Designed into workflow	Industrial dataset (offline)
[127]	Shaver logo printing	Products with uncertain score deferred to manual inspection	Empirical operating point	Designed into workflow	Industrial dataset (offline)
[131]	Semiconductor wafer maps	Defer low-confidence samples to experts	Empirical operating point	Designed into workflow	Benchmark (public)
[132]	Civil structures	PQT rejects outputs whose estimated accuracy is below a threshold; alerts opera...	Empirical operating point	Designed into workflow	Benchmark (public)
[54]	Industrial products	Three-way pass/review/reject thresholds from calibration data	Empirical operating point	Designed into workflow	Benchmark (public)
[145]	Dynamic industrial inspection via teleo...	VLM invoked below confidence threshold; human confirms safety-critical decisions	Empirical operating point	Designed into workflow	Industrial dataset (offline)
[20]	PV modules	Ambiguous cases escalated to experts	Empirical operating point	Evaluated with humans	Benchmark (public)
[67]	Industrial production lines	Two operating thresholds per model family; operator threshold variance measured	Empirical operating point	Evaluated with humans	Industrial dataset (offline)
[34]	Manufactured products	Threshold estimated from defect-free data only	Empirical operating point	None	Benchmark (public)
[17]	Railway catenary contact-wire supports	Image rejected as blurred if uncertainty > mean + 3 SD	Empirical operating point	None	Industrial dataset (offline)
[63]	Silicon wafer production	Novel if score > eta with P = alpha	Empirical operating point	None	Industrial dataset (offline)
[73]	Two-photon lithography microstructures	Batch flagged as novel if votes exceed calibrated threshold	Empirical operating point	None	Industrial dataset (offline)
[89]	Manufacturing inspection	Inputs below calibrated confidence threshold or out of distribution rejected fr...	Empirical operating point	None	Industrial dataset (offline)
[18]	Analog gauges in industrial plants	Proposed safeguards not validated	Empirical operating point	Recommended only	Industrial dataset (offline)
[19]	Concentrated solar power receivers	Risk-controlled threshold with abstention band to domain experts	Finite-sample	Designed into workflow	Deployed or operational
[142]	Industrial SEM inspection images and Ko...	Gate chosen on tuning data, certified on disjoint sample; review option	Finite-sample	Designed into workflow	Benchmark (public)
[121]	Industrial NDT components	Flag if p-value < 0.02	Finite-sample	None	Industrial dataset (offline)
[143]	Industrial surface inspection	Tolerance-limit threshold with P >= 1-delta	Finite-sample	None	Benchmark (public)
[148]	Electronics factory quality inspection	Humans approve or reject with AI score; no automated referral	None	Evaluated with humans	Human-subject experiment
[59]	Steel microstructure quality control	Thresholds confidential	None	Evaluated with humans	Industrial dataset (offline)
[27]	External pipe corrosion in chemical pla...	Personnel verify AI predictions	None	Evaluated with humans	Deployed or operational
[28]	External pipe corrosion in chemical pla...	Maintenance personnel verify outputs; high-risk-rank model planned	None	Evaluated with humans	Deployed or operational
[149]	Bridge overpasses	Inspector interprets AI detections via AR	None	Evaluated with humans	Field validation

chemical-plant piping [19], [22], [27], [28]. RQ2: robustness to shift dominates. Uncertainty is often estimated but seldom calibrated or linked to decisions, and sensor failure and grounding are emerging. RQ3: most studies stop at a score. Thresholds are rarely principled, only 4 provide finite-sample guarantees, and human review is mostly recommended rather than evaluated. RQ4: validation is mostly benchmark-based. Studies with natural shifts, field data or audits repeatedly show that benchmark performance overstates inspection reliability.

B. Cross-Cutting Patterns

Reading the map along the inspection chain (acquisition, model score, decision, human action) shows evidence thinning at each step. Acquisition and model-level robustness (R3, R6) are studied most. Score-level uncertainty (R1) is common but rarely calibrated. Decision-level guarantees (R2) and human-level evaluation (R5) are rare, and they are almost absent under natural shift and in energy settings. This interpretation is ours, derived from the coverage counts. It is consistent with the R8 studies showing that threshold-free metrics, saturated benchmarks and leakage hide decision-relevant errors [21], [34], [134], [143].

C. Implications for Energy and Industrial Inspection

For practitioners, the evidence argues for evaluating at the decision level: report false-alarm and miss rates at the operating point, calibrate on data resembling the target site, define referral policies and measure the review workload they create, and validate on natural shifts from the target fleet before relying on benchmark results. Guarantees reported in the literature assume exchangeable calibration data [142], [143]; they need recalibration or monitoring when sites, seasons or sensors change. Image-based findings should remain linked to physical verification (for example ultrasonic thickness or probability-of-detection studies [122], [126]) before maintenance or fitness-for-service conclusions are drawn.

D. Research Opportunities and Their Evidential Basis

Table VI lists opportunities that follow from documented patterns, each with its closest existing work. Sparse cells in the map are not proof of novelty: the 568 unretrieved reports and the single-index search may hide relevant studies.

E. Limitations of the Available Evidence

Most evidence comes from public benchmarks with controlled acquisition and synthetic shifts or contamination. Proprietary data limit reproducibility in industrial studies. Metrics are heterogeneous and mostly threshold-free, which prevents quantitative comparison. Several studies are preprints or appear in venues outside the field. They were charted as reported, with no quality weighting.

F. Limitations of the Review Process

First, the search used a single bibliographic index; seven of ten known primary studies were retrieved by the original search. Citation tracing has since been integrated but 701 reports still await title/abstract disposition and 66 screened reports await retrieval initiation. Second, full texts were retrieved only from open-access locations. 568 of 787 reports (largely IEEE, Elsevier and Springer) are not retrieved and not charted from full text, which may bias the map toward open preprints and journals. An exploratory summary of screening rationales covers 364 unretrieved reports with a captured abstract; 204 others lack captured abstracts. This is a report-level retrieval-gap inventory, not independent full-text charting or a test of retrieval bias. Reliability dimensions were extracted from prior screening rationales; energy-title keywords are not interchangeable with the full-text context coding. It cannot establish that missing reports would leave the evidence map unchanged. Third, selection and charting were performed by an AI assistant under the first author's delegation, with one reviewer per decision. The current local audit file contains no populated auditor decisions; the author reports that completed entries exist, but their location is being reconciled. No agreement statistic or completed audit is claimed here. Fourth, full texts in languages other than English were excluded (3 reports). Fifth, the window closes in October 2026, during a period of fast growth; recent preprints may change after peer review.

[Author input required: update this subsection after the human audit, any manual retrieval of unretrieved reports and citation tracing.]

VII. Conclusion

Within the 157 studies provisionally assessed to date, reliability research on AI-based visual inspection has expanded rapidly but remains centred on model-level robustness evaluated on public manufacturing benchmarks. Decision-level evidence is scarce: few studies give guarantees, measure referral workload, test real sensor failures or validate in the field, especially for energy assets. Where realistic evaluation is done, it repeatedly shows that benchmark scores overstate inspection reliability. These observations motivate decision-level evaluation under natural shift, with explicit operating points, referral policies and field validation. They remain provisional until outstanding search additions and audit reconciliation are resolved.

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[Author input required: acknowledgments, if any.]

Declarations and Research Artifacts

Data and Code Availability

Protocol, search queries and raw responses, decision records, linkage, retrieval logs, charting data, and scripts are available in the project repository (https://github.com/Engr-Daniel/trustworthy-ai-energy).

[Author input required: archived release DOI or commit.]

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Funding and Conflicts of Interest

This research received no external funding.

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Author Contributions and AI Assistance

D. Oyewale conceived the review, set the protocol, assessed the calibration sample and delegated selection and charting. An AI assistant (Claude, Anthropic) performed the earlier searches, screening, full-text eligibility, charting and drafting under that delegation, with recorded rationales and locators. On 2 November, an OpenAI assistant reconciled saved decision packets, checked derived counts and revised the provisional draft. This reconciliation was not an independent second eligibility review.

[Author input required: confirm contributions and the human verification performed; the AI assistant is not an author.]

Appendix A
Complete Search Strategies

Exact OpenAlex query strings, fields, filters, pagination and per-page checksums are in `review-paper/search-strategy/openalex-queries.json` and `runs/OPENALEX-20261022-0[12]/search_manifest.json`; failed and superseded runs (`OPENALEX-20261016-01–03`), the Crossref pilot and seed metadata are in the same location, and `corpus/search-runs.csv` lists every run.

Appendix B
Charting Codebook and Decision Rules

The charting form (versions 0.1 and 0.2), screening notes 1–14, reason codes and the R1–R8 definitions are in `review-paper/charting/charting-form.md`, `screening/ta-ai-20261022/screening-notes.md` and `screening/reason-codes.csv`; charted values with page locators are in `charting/charting.csv`.

TABLE VI
EVIDENCE-GROUNDED RESEARCH OPPORTUNITIES.

Observed pattern	Closest work	Question	Required data/evaluation	Qualification
Finite-sample guarantees are rare and assume exchangeability	[19], [121], [142], [143]	Do risk-controlled thresholds keep their coverage under natural site and season shift in energy inspection?	Multi-site data with natural shift; coverage and false-alarm audit over time	Guarantees may need shift-aware or online recalibration
Referral is recommended more than evaluated	[14], [26], [54], [148]	What referral policy minimises missed defects for a fixed inspector workload?	Inspector studies with measured workload, time and error	Human studies are costly; simulation is weaker evidence
Sensor failure tested mostly synthetically	[30], [31], [35], [88]	How do RGB–thermal UAV inspection models behave with real sensor degradation?	Field data with logged sensor faults; missing-modality tests	Needs operator data access
VLM grounding is weak and rarely validated against standards	[20], [32], [33], [147]	Can grounded VLM reports be verified against inspection standards and expert findings?	Expert-annotated reports; hallucination and faithfulness metrics	Rapidly moving model base
Benchmarks hide decision errors	[21], [34], [126], [134]	Which evaluation protocol predicts field reliability of visual inspection?	Leakage-free splits, operating-point metrics, PoD, field follow-up	Requires shared field datasets

APPENDIX C
REPORTING CHECKLIST AND SUPPLEMENTARY EVIDENCE

[Author input required: complete the PRISMA-ScR checklist with final page numbers after typesetting.]

Full inclusion and exclusion records with rationales are in `screening/confirmed-decisions.csv`; reports not retrieved are listed in `fulltext/not-retrieved.csv`.

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